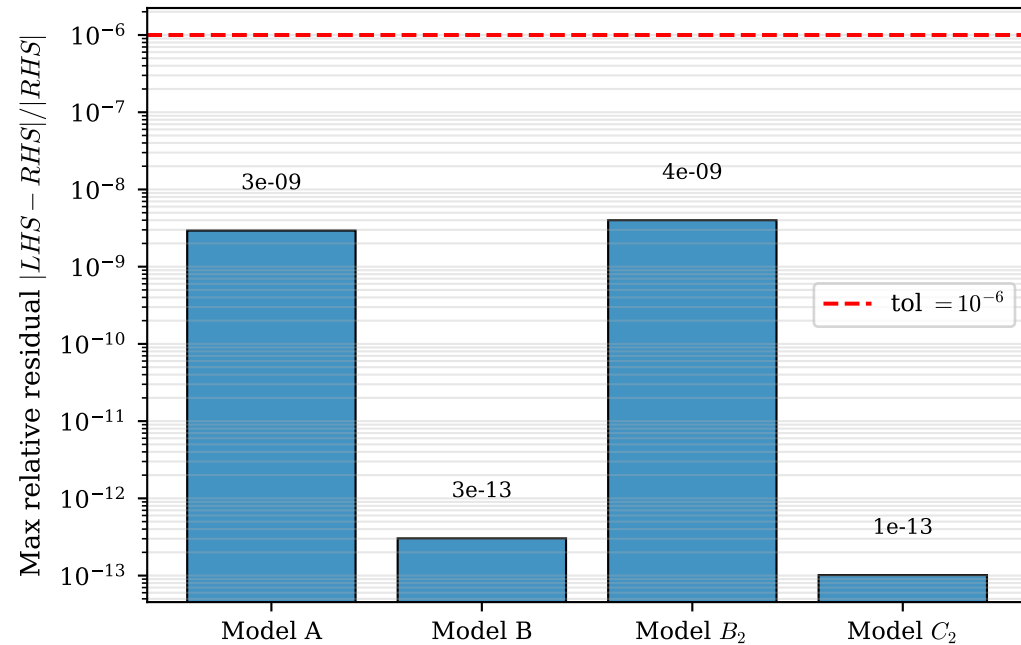


Structural lemmas: residual of both sides of each functional equation

Lemma 1 — fundamental PGF equation



Lemma 2 — PPGF dynamics on $\tilde{S}$

